Semiconductor Devices - Lecture Notes

1. Semiconductor Fundamentals

Silicon has four valence electrons. Doping with a pentavalent impurity (phosphorus) gives n-type material with electrons as majority carriers; a trivalent impurity (boron) gives p-type material with holes as majority carriers. The intrinsic carrier concentration of silicon at 300 K is about 1.5e10 per cubic centimetre.

2. The PN Junction Diode

A forward-biased diode conducts above the barrier potential, about 0.7 volts for silicon and 0.3 volts for germanium. The diode current is I = Is*(exp(V/(n*Vt)) - 1), where Vt is the thermal voltage, approximately 26 mV at room temperature, and n is the ideality factor (close to 1 for silicon).

Reverse bias produces only a small leakage current. A Zener diode operates in reverse breakdown and is used for voltage regulation; an LED is a forward-biased diode that emits light.

3. Bipolar Junction Transistor

A BJT has three terminals: emitter, base and collector. In the active region the emitter-base junction is forward biased and the collector-base junction is reverse biased. The current gain beta (h_FE) relates collector current to base current: Ic = beta*Ib. The relation between alpha and beta is alpha = beta/(beta + 1) and beta = alpha/(1 - alpha).

In saturation the transistor behaves as a closed switch with Vce about 0.2 V; in cut-off it behaves as an open switch.

4. Field Effect Transistors

A MOSFET is a voltage-controlled device. In an n-channel enhancement MOSFET, conduction begins when the gate-source voltage exceeds the threshold voltage Vth. In the saturation region the drain current is Id = 0.5*kn*(Vgs - Vth)^2. The transconductance is gm = 2*Id/(Vgs - Vth).

MOSFETs have very high input impedance (gates draw almost no current), low power dissipation and are the basis of CMOS logic.